Measurement 221 (2023) 113508

Contents lists available at ScienceDirect

Measurement

journal homepage: www.elsevier.com/locate/measurement

Simultaneous calibration of hand-eye and kinematics for industrial robot using line-structured light sensor

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ARTICLE INFO

Keywords:
Industrial robot
Simultaneous calibration
Hand-eye error
Kinematic error
Distance constraint

ABSTRACT

For visually guided industrial robots, it is necessary to calibrate the robot's hand-eye relationship to accurately guide the robot's operations with the measurement data of the visual coordinate system. In most cases, the calibration process of hand-eye error does not consider the impact of kinematic errors, which accumulates the error in the calibration process and reduces the calibration accuracy. To enhance the calibration accuracy of industrial robots, a simultaneous calibration method of robot hand-eye relationship and kinematics using a line-structured light sensor is proposed. A simultaneous calibration model is established by using the differential kinematics principle. Afterward, the line-structured light sensor and two standard balls with known center distances are used in the calibration and accuracy verification experiments. After calibration and compensation with the proposed method, the average distance error of the robot declined from 3.967 mm to 1.001 mm. The proposed simultaneous calibration method for hand-eye relationship and kinematics dramatically improves the robot's positioning accuracy.

1. Introduction

Industrial robots are intelligent devices with the advantages of automation, high adaptability, and controllability, which are widely used in manufacturing, aerospace, medical, and service fields [1]. Initially, robots were primarily used in automotive manufacturing processes that required low positioning accuracies, such as loading and unloading, handling, welding, and painting [2–4]. With the continuous development of manufacturing and control technology, the application range of industrial robots has gradually expanded to the precision manufacturing industry [5–6]. One of the industry's issues is the low positioning accuracy of industrial robots, which is the main issue preventing robot development for precision manufacturing [7]. Based on structural and external factors, the elements that affect robot positioning accuracy can be divided into kinematic errors and non-kinematic errors. According to a study, kinematic errors account for roughly 80% of total errors [8]. Therefore, kinematic errors in industrial robots must be corrected to meet the requirements of their industrial applications.

Error calibration, a straightforward and efficient method to increase the positioning accuracy of robots, can significantly improve the accuracy of robot through four steps: error modeling, measurement, parameter identification, and compensation [9–11]. In response to the large amount of uncertainty in the actual work of robots, researchers have also proposed effective methods to improve robot positioning accuracy. Santolariaet al. [12] proposed an uncertainty representation method for the kinematic calibration based on the probability distribution propagation calculation. Furthermore, Yang et al. [13] proposed a reliability analysis method for the positioning accuracy based on a non-probabilistic time-dependent model, which is suitable for the positioning accuracy analysis of industrial robots with lines, curves, and spiral trajectories. Error measurement is a prerequisite for error compensation. The position of the robot's end can be directly obtained by high-precision measurement devices such as laser trackers and laser scanners [14–15]. High-precision measurement devices are not widely utilized by factories because they are expensive and will cause data transmission delays during the online process. As a result of the constant advancement of vision technology, camera-based indirect measurement devices have been implemented in the robot pose measurement process. Zhang et al. [16] measured the robot's attitude error by using the monocular camera to identify the pixel points on the plane target to determine the robot's final position. Kuo et al. [17] established a monocular vision system, identified the locations of four feature points

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https://doi.org/10.1016/j.measurement.2023.113508
Received 28 June 2023; Received in revised form 11 August 2023; Accepted 30 August 2023
Available online 1 September 2023

at various joints of the robot using pinhole imaging technology, and then used image processing to obtain the robot's position and posture. Measurement using vision devices is more widespread due to its lower cost, despite its lower accuracy than that of high-precision measurement devices.

In order to use the vision system to guide robot perception and operation, it is necessary to accurately obtain the "hand-eye relationship" matrix between the robot end and the vision system. According to Shiu and Ahmad [18], it is comparable to the issue of solving the rotation matrix **R** and translation vector **t**, as well as the problem of solving the nonlinear homogeneous equation **AX** = **BX**. The rotation matrix **R** and the translation vector **t** are estimated separately by numerical solution, which will result in a transfer of the rotation matrix's estimation error to the position vector's estimation outcome [19]. Researchers proposed several "simultaneous solutions" to simultaneously obtain the **R** and **t** to get rid of the error transference error [20–22].

Although numerous calibration methods have been proposed to increase the accuracy of robots, the majority of them handle hand-eye calibration and kinematic error calibration tasks separately. Existing calibration methods have not accounted for the coupling effect between hand-eye error and kinematic error, resulting in error accumulation. To solve the aforementioned issues, this work proposes a solution for the simultaneous calibration of hand-eye and kinematics based on a line-structured light sensor. The separation method is used to obtain the initial values of the hand-eye parameters, which are then brought into the simultaneous calibration model to calibrate the error parameters with the Levenberg-Marquardt (LM) algorithm. The direct compensation method and iterative compensation method are used in the off-line error compensation process for the robot. And the effectiveness and correctness of the method are verified by adopting the KUKA robot as the experimental object.

2. Error simultaneous calibration modeling

2.1. Kinematic model

The typical Denavit-Hartenberg (D-H) convention is used to model the kinematics of a KUKA robot that has n joints and $n+1$ linkages. The transformation matrix ${}^{i-1}_{i}\boldsymbol{T}$ between two consecutive joints can be described by the following equation.

$$\begin{aligned}{}^{i-1}_{i}\boldsymbol{T} &= \mathrm{Rot}(x_{i-1},\alpha_{i-1})\mathrm{Trans}(x_{i-1},a_{i-1})\mathrm{Rot}(z_i,\theta_i)\mathrm{Trans}(z_i,d_i)\\ &= \begin{bmatrix} \cos\theta_i & -\sin\theta_i & 0 & a_{i-1}\\ \sin\theta_i\cos\alpha_{i-1} & \cos\theta_i\cos\alpha_{i-1} & -\sin\alpha_{i-1} & -d_i\sin\alpha_{i-1}\\ \sin\theta_i\sin\alpha_{i-1} & \cos\theta_i\sin\alpha_{i-1} & \cos\alpha_{i-1} & d_i\cos\alpha_{i-1}\\ 0 & 0 & 0 & 1\end{bmatrix}\end{aligned} \tag{1}$$

where α_{i-1} is the connecting rod deflection angle, a_{i-1} is the connecting rod length, θ_i is the joint angle, and d_i is the rod offset.

The base coordinate system and joint coordinate system of the KUKA KR210 robot are defined in Fig. 1. Table 1 displays the nominal parameter values for the D-H model of the KUKA KR210 robot.

A consecutive multiplication of the transformation matrix of each joint can be used to define the transformation from the robot base to its end coordinate system, as in Eq. (2).

$${}^{B}_{n}\boldsymbol{T}(\theta) = {}^{B}_{1}\boldsymbol{T}(\theta_1)\,{}^{1}_{2}\boldsymbol{T}(\theta_2)\cdots{}^{n-1}_{n}\boldsymbol{T}(\theta_n) \tag{2}$$

2.2. Simultaneous calibration model

Assuming that the theoretical and actual values of the robot's kinematic parameters differ by a tiny margin, which is denoted as, respectively: the error in the connecting rod length Δa_{i-1}, the error in the connecting rod deflection angle $\Delta\alpha_{i-1}$, the error in the rod offset Δd_i, and the error in the joint angle $\Delta\theta_i$. The result of fully differentiating the positional matrix ${}^{i-1}_{i}\boldsymbol{T}$ is:

$$\begin{aligned} d_i^{i-1}\ \boldsymbol{T} &= \frac{\partial_i^{i-1}\boldsymbol{T}}{\partial a_{i-1}}\Delta a_{i-1} + \frac{\partial_i^{i-1}\boldsymbol{T}}{\partial \alpha_{i-1}}\Delta\alpha_{i-1} + \frac{\partial_i^{i-1}\boldsymbol{T}}{\partial d_i}\Delta d_i + \frac{\partial_i^{i-1}\boldsymbol{T}}{\partial\theta_i}\Delta\theta_i\\ &=_i^{i-1}\ \boldsymbol{T}\,{}^{i}\Delta \end{aligned} \tag{3}$$

where ${}^{i}\Delta$ is the error operator induced by the kinematic error of the connecting rod i.

Splitting the matrix ${}^{i}\Delta$ produces the sum of error operators consist-

Table 1
Nominal D-H parameters of the KUKA KR210 robot.

Joint(i)	$a_{(i-1)}$/mm	$\alpha_{(i-1)}$/°	d_i/mm	θ_i/°
1	0	0	675	θ_1
2	350	−90	0	θ_2-90
3	1150	0	0	θ_3
4	41	−90	1200	θ_4
5	0	90	0	θ_5
6	0	−90	215	θ_6

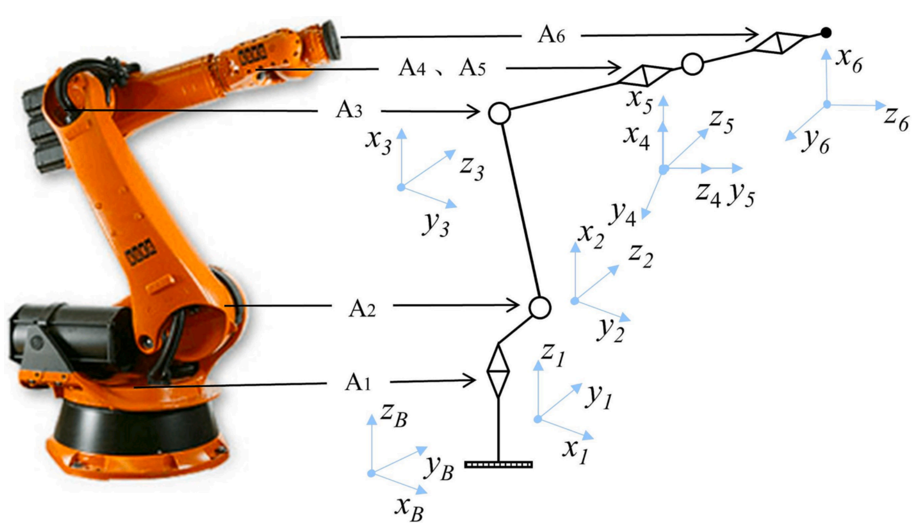

Fig. 1. Coordinate system of KUKA robot.

ing of four error terms(Δa_{i-1}, $\Delta\alpha_{i-1}$, Δd_i, $\Delta\theta_i$).

$$ {}^{i}\boldsymbol{\Delta} = \boldsymbol{T}_{a_{i-1}}\Delta a_{i-1} + \boldsymbol{T}_{\alpha_{i-1}}\Delta\alpha_{i-1} + \boldsymbol{T}_{d_i}\Delta d_i + \boldsymbol{T}_{\theta_i}\Delta\theta_i \tag{4}$$

where $\boldsymbol{T}_{a_{i-1}}$, $\boldsymbol{T}_{\alpha_{i-1}}$, $\boldsymbol{T}_{d_i}$, and $\boldsymbol{T}_{\theta_i}$ are the coefficient matrices of Δa_{i-1}, $\Delta\alpha_{i-1}$, Δd_i, and $\Delta\theta_i$ respectively.

Combining Eq. (2) and Eq. (3), the total error of the end flange center caused by the kinematic error of each joint is:

$$\begin{aligned} {}_{m}^{B}\boldsymbol{T} + d_{m}^{B}\boldsymbol{T} &= \prod_{i=1}^{m}\left({}_{i}^{i-1}\boldsymbol{T} + d_{i}^{i-1}\boldsymbol{T}\right) \\ &= \left({}_{1}^{B}\boldsymbol{T} + d_{1}^{B}\boldsymbol{T}\right)\left({}_{2}^{1}\boldsymbol{T} + d_{2}^{1}\boldsymbol{T}\right)\cdots\left({}_{m}^{m-1}\boldsymbol{T} + d_{m}^{m-1}\boldsymbol{T}\right) \end{aligned} \tag{5}$$

Expanding Eq. (5) and neglecting the higher-order differential error terms leads to Eq. (6).

$$ {}_{m}^{B}\boldsymbol{T} + d_{m}^{B}\boldsymbol{T} = {}_{m}^{B}\boldsymbol{T} + \sum_{i=0}^{m-1}\left({}_{i}^{B}\boldsymbol{T}\left(d_{i+1}^{i}\boldsymbol{T}\right){}_{m}^{i+1}\boldsymbol{T}\right) \tag{6}$$

Substituting Eq. (3) into Eq. (6), the error matrix $d_{m}^{B}\boldsymbol{T}$ of the end $\{m\}$ due to the kinematic error of each linkage is:

$$\begin{aligned} d_{m}^{B}\boldsymbol{T} &= \sum_{i=1}^{m}\left({}_{i}^{B}\boldsymbol{T}\left({}^{i}\boldsymbol{\Delta}\right){}_{m}^{i}\boldsymbol{T}\right) \\ &= \sum_{i=1}^{m}\left({}_{i}^{B}\boldsymbol{T}\boldsymbol{T}_{a_{i-1}}{}_{m}^{i}\boldsymbol{T}\Delta a_{i-1} + {}_{i}^{B}\boldsymbol{T}\boldsymbol{T}_{\alpha_{i-1}}{}_{m}^{i}\boldsymbol{T}\Delta\alpha_{i-1} + {}_{i}^{B}\boldsymbol{T}\boldsymbol{T}_{d_i}{}_{m}^{i}\boldsymbol{T}\Delta d_i + {}_{i}^{B}\boldsymbol{T}\boldsymbol{T}_{\theta_i}{}_{m}^{i}\boldsymbol{T}\Delta\theta\right) \end{aligned} \tag{7}$$

The position error vector ${}^{B}\boldsymbol{E}_{q}^{m}$ at the end $\{m\}$ of the robot due to the kinematic error of each linkage can be obtained by performing the differential operator inverse operation.

$$\begin{aligned} {}^{B}\boldsymbol{E}_{q}^{m} &= \left(d_{m}^{B}\boldsymbol{T}\left({}_{m}^{B}\boldsymbol{T}\right)^{-1}\right)' \\ &= \left[\boldsymbol{k}_{q1}^{m} \quad \cdots \quad \boldsymbol{k}_{qm}^{m}\right]\begin{bmatrix}\Delta\boldsymbol{q}_1 \\ \vdots \\ \Delta\boldsymbol{q}_m\end{bmatrix} \\ &= \boldsymbol{k}_{q}^{m}\Delta\boldsymbol{q}_{\theta} \end{aligned} \tag{8}$$

where $\boldsymbol{k}_{qi}^{m}$ is the error coefficient matrix corresponding to the positional error of the robot end coordinate system $\{m\}$ due to the deviation of linkage *i*. $\boldsymbol{k}_{q}^{m}$ is the total error coefficient matrix of the robot end coordinate system $\{m\}$ due to the robot kinematic error. $\Delta\boldsymbol{q}_i$ is the vector of the kinematic error parameter corresponding to joint *i*, and $\Delta\boldsymbol{q}_{\theta}$ is the column vector composed of the errors of the robot's each kinematic parameter.

In the eye-in-hand configuration, the robot establishes connections and communication with the external environment through the vision sensor. When the hand-eye error exists, it is necessary to establish the representation ${}^{B}\boldsymbol{E}_{q}^{s}$ of the robot vision system error due to the kinematic error of each linkage in the base coordinate system $\{B\}$. Since the vision sensor is anchored to the robot's end flange, it can be obtained by the rigid body error differential transformation and equation (8).

$$\begin{aligned} {}^{B}\boldsymbol{E}_{q}^{s} &= \begin{bmatrix} {}_{B}^{s}\boldsymbol{R} & -{}_{B}^{s}\boldsymbol{R}\left[{}^{s}\boldsymbol{t}_{B}\right] \\ 0 & {}_{B}^{s}\boldsymbol{R}\end{bmatrix}{}^{B}\boldsymbol{E}_{q}^{m} \\ &= \left[\boldsymbol{J}_{q1}^{s} \quad \cdots \quad \boldsymbol{J}_{qm}^{s}\right]\begin{bmatrix}\Delta\boldsymbol{q}_1 \\ \vdots \\ \Delta\boldsymbol{q}_m\end{bmatrix} \\ &= \boldsymbol{J}_{q}^{s}\Delta\boldsymbol{q}_{\theta} \end{aligned} \tag{9}$$

where ${}_{B}^{s}\boldsymbol{R}$ is the rotation matrix, ${}^{s}t_{B}$ is the position vector, $\boldsymbol{J}_{qi}^{s}$ is the Jacobian matrix corresponding to $\Delta\boldsymbol{q}_i$, and $\boldsymbol{J}_{q}^{s}$ is the representation of the total Jacobian matrix of the linkage kinematic error in the visual coordinate system $\{s\}$.

To construct a unified expression for the robot hand-eye error and kinematic error, it is required to further solve for the representation ${}^{B}\boldsymbol{E}_{s}$ of the robot vision system error due to the hand-eye error in the base coordinate system $\{B\}$.

Assuming that the hand-eye relationship error is $\Delta\boldsymbol{q}_s = \left[d_s^T \quad \delta_s^T\right] = \left[{}^{s}d_x \quad {}^{s}d_y \quad {}^{s}d_z \quad {}^{s}\delta_x \quad {}^{s}\delta_y \quad {}^{s}\delta_z\right]^T$. According to the basic principle of differential transformation, ${}^{B}\boldsymbol{E}_{s}$ can be obtained.

$$\begin{aligned} {}^{B}\boldsymbol{E}_{s} &= \left[\boldsymbol{d}_{B}^{T} \quad \boldsymbol{\delta}_{B}^{T}\right]^{T} \\ &= \boldsymbol{J}_{s}\Delta\boldsymbol{q}_{s} \end{aligned} \tag{10}$$

As a result, the overall error induced by hand-eye error and kinematic error observed in the robot's base coordinate system $\{B\}$ is the sum of ${}^{B}\boldsymbol{E}_{s}$ and ${}^{B}\boldsymbol{E}_{q}^{s}$.

$$\begin{aligned} {}^{B}\boldsymbol{E} &= {}^{B}\boldsymbol{E}_{q}^{s} + {}^{B}\boldsymbol{E}_{s} \\ &= \left[\boldsymbol{J}_{q1} \quad \cdots \quad \boldsymbol{J}_{qm} \quad \boldsymbol{J}_{s}\right]\begin{bmatrix}\Delta\boldsymbol{q}_1 \\ \vdots \\ \Delta\boldsymbol{q}_m \\ \Delta\boldsymbol{q}_s\end{bmatrix} \\ &= \boldsymbol{J}\Delta\boldsymbol{q} \end{aligned} \tag{11}$$

where ${}^{B}\boldsymbol{E}$ is the overall error vector of the robot vision system in the base coordinate system. $\boldsymbol{J}$ represents the total Jacobian matrix influenced by the robot's kinematic parameters and the hand-eye interaction. $\Delta\boldsymbol{q}$ is composed of the error vector $\Delta\boldsymbol{q}_i$ for each robot linkage and the hand-eye error vector Δq_s. Each linkage error vector $\Delta\boldsymbol{q}_i$ has four parameters, and the hand-eye error vector Δq_s has six.

The unified model of robot hand-eye and kinematic errors is the foundation for identifying kinematic parameters. The unified error model reflects the mapping relationship between robot position deviation and errors for hand-eye and kinematic parameters.

3. Error identification based on distance constraint

Based on the previous chapter's robot error simultaneous calibration model, the least squares objective equation is established in this chapter, and the LM algorithm is used to calculate hand-eye and kinematic parameter deviations. The least squares error value is used as the initial value of the LM algorithm, which can achieve the optimal error value faster. Since there is no uniform sampling space for the classic point constraint measurement, this study collects error data based on the distance constraint. Two standard balls are separated by a specified distance, and then the error data are measured by a line-structured light sensor. In order to adapt to the working stroke of the subsequent weld seams grinding, the spacing distance is taken as 140 mm.

3.1. Hand-eye initial calibration

The robot hand-eye relationship is referred to the relative positional relationship between the vision measurement system and the robot end coordinate system. A robot measurement system is constructed that is based on a line-structured light sensor, and the robot's base coordinate system, each joint coordinate system, end coordinate system, and measurement coordinate system is defined, as shown in Fig. 2. The transformation matrix between the robot end coordinate system $\{m\}$ and the visual coordinate system $\{s\}$ is the robot hand-eye relationship to be calibrated.

The hand-eye calibration technique utilized in this study is an extension of the fixed-point technique, which establishes the point constraint equations using the constant location of the standard ball's center ${}^{s}P$ in the base coordinate system. The standard ball utilized is a high precision ball with a known diameter. The cross section of the sphere produced by laser irradiation in any direction is circular, and the line connecting the cross-section's center to the sphere's center is

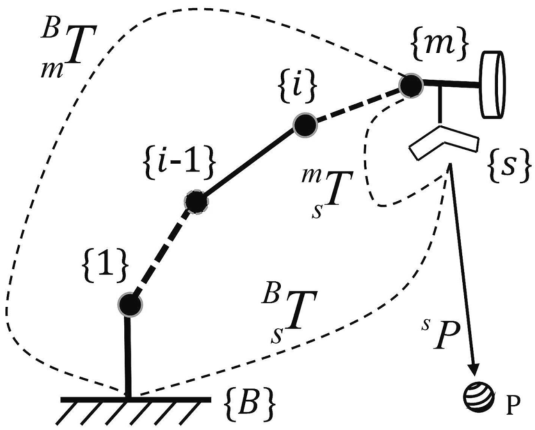

Fig. 2. Line-structured light vision measurement system.

perpendicular to the tangent plane. Based on this kinematic relationship, the coordinates of the standard ball's center in the visual coordinate system can be calculated. The robot's base coordinate system is fixed, and the coordinates of the standard ball's center under the base coordinate system are invariant values, so the hand-eye relationship matrix can be obtained.

For the line-structured light sensor, only two dimensions of x and z can be output, with a default of 0 in the y-direction ($y = 0$). When a standard ball is scanned with a laser line, a circular arc is created on which the x and z data of each point from the vision sensor measurement system can be obtained. The radius and center coordinates of the arc can be obtained from the measured two-dimensional data. The spherical center coordinates of the standard ball can be calculated from Eq. (12).

$$\begin{cases} x_R = x_r \\ y_R = \pm\sqrt{R^2 - r^2} \\ z_R = z_r \end{cases} \tag{12}$$

where R is the radius of the standard ball, x_r and z_r are the two-dimensional coordinate values of the arc, and r is the radius of the arc.

The positive or negative values of y_R need to be determined by the relative position of the center of the standard ball and the laser line tangent. However, it might be challenging to precisely distinguish the positive or negative y-value when the laser line is close to the center of the ball. This may lead to inaccurate results of hand-eye calibration and kinematics calibration. The following solution can be used to precisely identify the positive or negative y-values.

As shown in Fig. 3, the robot is controlled to approach the standard ball with any attitude. Near the ball's center, a measurement is taken and recorded as an arc①, and the 2D data collected by the measuring equipment is recorded to calculate the arc radius R_1. Then, taking the arc ①as the center, the robot is controlled to translate Δd to the left and right, respectively, obtaining arcs ②and ③with radius of R_2 and R_3. As shown in Fig. 3a, if $R_1 > R_2 > R_3$, y_1 and y_2 are negative, and y_3 is positive, and if $R_1 > R_3 > R_2$, y_1 is negative, and y_2 and y_3 are positive. In Fig. 3b, $R_2 > R_1 > R_3$, then y_1 and y_3 are positive, y_2 can not be determined. In Fig. 3c, $R_3 > R_1 > R_2$, then y_1 and y_2 are negative, y_3 can not be determined. The proposed judgment scheme can judge the positive or negative value of y at two positions at one time, avoiding the error of artificial judgment of the positive or negative value of the y coordinate.

Assuming that there is a point P which has coordinates ${}^{B}P$ under the robot's base coordinate system $\{B\}$ and ${}^{s}P$ under the visual coordinate system $\{s\}$, the relational equation for converting the vision system's measurements to the robot's base coordinate system is as follows:

$${}^{B}\widetilde{\boldsymbol{P}} = {}^{B}_{m}\boldsymbol{T}_i \times {}^{m}_{s}\boldsymbol{T} \times {}^{s}\widetilde{\boldsymbol{P}}_i \tag{13}$$

where ${}^{B}_{m}\boldsymbol{T}_i$ is the relationship matrix from the robot's end coordinate system $\{m\}$ to the base coordinate system $\{B\}$, the hand-eye relationship matrix ${}^{m}_{s}\boldsymbol{T}$ is a fixed value, and ${}^{s}\widetilde{\boldsymbol{P}}_i$ is the position vector of the standard ball under the visual coordinate system $\{s\}$ during the i_{th} measurement. ${}^{B}\widetilde{\boldsymbol{P}} = \begin{bmatrix} x_{P_B} & y_{P_B} & z_{P_B} & 1 \end{bmatrix}^{\mathrm{T}}$, ${}^{s}\widetilde{\boldsymbol{P}} = \begin{bmatrix} x_{P_s} & y_{P_s} & z_{P_s} & 1 \end{bmatrix}^{\mathrm{T}}$.

The robot is manipulated to measure the standard ball fixed in the world coordinate system in different positions, and the following relationship is given.

$$\begin{bmatrix} {}^{B}_{m}\boldsymbol{R}_i & {}^{B}_{m}\boldsymbol{t}_i \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} {}^{m}_{s}\boldsymbol{R} & {}^{m}_{s}\boldsymbol{t} \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} \boldsymbol{p}_i \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{B}_{m}\boldsymbol{R}_{i+1} & {}^{B}_{m}\boldsymbol{t}_{i+1} \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} {}^{m}_{s}\boldsymbol{R} & {}^{m}_{s}\boldsymbol{t} \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} \boldsymbol{p}_{i+1} \\ 1 \end{bmatrix} \tag{14}$$

First, the robot is controlled to measure the fixed standard ball in a way that only translates but not rotates. The rotation matrix ${}^{B}_{m}\boldsymbol{R}$ in the equation does not change because the robot does not rotate but only moves to complete two measurements of the fixed point, that is ${}^{B}_{\mathrm{m}}\boldsymbol{R}_i = {}^{B}_{\mathrm{m}}\boldsymbol{R}_{i+1} = {}^{B}_{\mathrm{m}}\boldsymbol{R}$. Therefore, from Eq. (14), it becomes that:

$${}^{B}_{m}\boldsymbol{R}^{-1}\left({}^{B}_{m}\boldsymbol{t}_{i+1} - {}^{B}_{m}\boldsymbol{t}_i\right) = {}^{m}_{s}\boldsymbol{R}\left({}^{s}\boldsymbol{p}_i - {}^{s}\boldsymbol{p}_{i+1}\right) \tag{15}$$

The robot is controlled to perform n measurements on the fixed standard ball to obtain n-1 equations and then a singular value decomposition (SVD) of Eq. (15) to calculate the rotation matrix $\mathbf{R}$. The solved rotation matrix $\mathbf{R}$ is brought into Eq. (14).

$$\left({}^{B}_{m}\boldsymbol{R}_i - {}^{B}_{m}\boldsymbol{R}_{i+1}\right)\cdot {}^{m}_{s}\boldsymbol{t} = \left({}^{B}_{m}\boldsymbol{t}_{i+1} - {}^{B}_{m}\boldsymbol{t}_i\right) + {}^{B}_{m}\boldsymbol{R}_{i+1}\cdot {}^{m}_{s}\boldsymbol{R}\cdot {}^{s}\boldsymbol{p}_{i+1} - {}^{B}_{m}\boldsymbol{R}_i\cdot {}^{m}_{s}\boldsymbol{R}\cdot {}^{s}\boldsymbol{p}_i \tag{16}$$

The robot is then programmed to take numerous measurements of the fixed standard ball at any position and attitude to get the position vector ${}^{m}_{s}\boldsymbol{t}$. The Eq. (16) can be simplified as Eq. (17), and the value of the position vector ${}^{m}_{s}\boldsymbol{t}$ can be calculated by solving it with the least squares method.

$$\begin{bmatrix} \boldsymbol{A}_1 \\ \vdots \\ \boldsymbol{A}_n \end{bmatrix}_{3n\times 3} \begin{bmatrix} x \\ y \\ z \end{bmatrix}_{3\times 1} = \begin{bmatrix} \boldsymbol{b}_1 \\ \vdots \\ \boldsymbol{b}_n \end{bmatrix}_{3n\times 1} \tag{17}$$

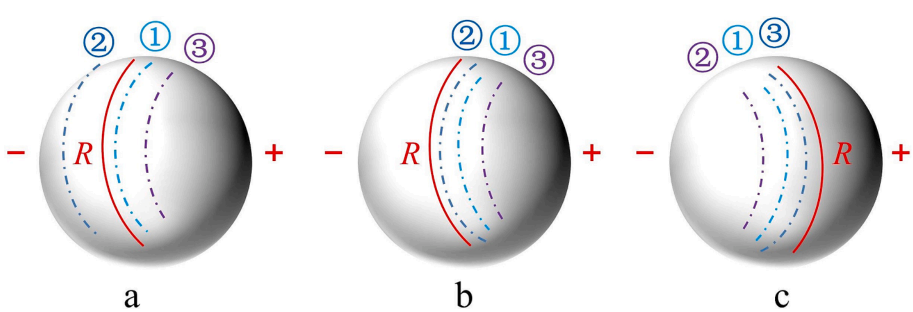

Fig. 3. Schematic diagram of laser line distribution. (a) case 1, (b) case 2, (c) case 3.

3.2. Error model based on point constraint

The relationship between the coordinates $\boldsymbol{p}_B$ of any point P under the base coordinate system $\{B\}$ and the coordinates $\boldsymbol{p}_s$ under the visual coordinate system $\{s\}$ is shown in Eq. (13). Assuming that the robot kinematic error and the hand-eye error are known, Eq. (18) describes the actual coordinate value of the point P.

$$\begin{cases} {}^B\widetilde{\boldsymbol{P}} = \left({}^B_s\boldsymbol{T} + {}^s\boldsymbol{\Delta}^B_s\boldsymbol{T}\right)\times {}^s\widetilde{\boldsymbol{P}} \\ \begin{bmatrix} \boldsymbol{p}_B+\Delta \boldsymbol{p}_B \\ 1\end{bmatrix} = \left(\begin{bmatrix} {}^B_s\boldsymbol{R} & {}^B_s\boldsymbol{t} \\ 0 & 1\end{bmatrix} + \begin{bmatrix}[{}^s\boldsymbol{\delta}] & {}^s\boldsymbol{d} \\ 0 & 1\end{bmatrix}\begin{bmatrix}{}^B_s\boldsymbol{R} & {}^B_s\boldsymbol{t}\\ 0 & 1\end{bmatrix}\right)\times\begin{bmatrix}\boldsymbol{p}_s\\ 1\end{bmatrix}\end{cases} \tag{18}$$

where ${}^s\boldsymbol{\Delta}$ is the differential operator corresponding to $d^m_s\boldsymbol{T}$, ${}^s\boldsymbol{d}$ and ${}^s\boldsymbol{\delta}$ are the differential translation vector and the differential rotation vector, respectively.

Eq. (18) is further expanded to obtain Eq. (19).

$$\begin{aligned} \boldsymbol{p}_B + \Delta\boldsymbol{p}_B &= \left([{}^s\boldsymbol{\delta}]{}^B_s\boldsymbol{R} + {}^B_s\boldsymbol{R}\right)\boldsymbol{p}_s + [{}^s\boldsymbol{\delta}]{}^B_s\boldsymbol{t} + {}^B_s\boldsymbol{t} + {}^s\boldsymbol{d} \\ &= \begin{bmatrix}\boldsymbol{I}_{3\times3} & [{}^B_s\boldsymbol{R}\boldsymbol{p}_s] + [{}^B_s\boldsymbol{t}]\end{bmatrix}\begin{bmatrix}{}^s\boldsymbol{d}\\ {}^s\boldsymbol{\delta}\end{bmatrix} + {}^B_s\boldsymbol{R}\boldsymbol{p}_s + {}^B_s\boldsymbol{t}\end{aligned} \tag{19}$$

Putting $\boldsymbol{p}_B = {}^B_s\boldsymbol{R}\boldsymbol{p}_s + {}^B_s\boldsymbol{t}$ into Eq. (19) and removing $\boldsymbol{p}_B$ from both sides produces the error value of the measurement point P, as in Eq. (20).

$$\begin{aligned}\Delta\boldsymbol{p}_B &= \begin{bmatrix}\boldsymbol{I}_{3\times3} & [{}^B_s\boldsymbol{R}\boldsymbol{p}_s] + [{}^B_s\boldsymbol{t}]\end{bmatrix}\boldsymbol{J}\Delta\boldsymbol{q} \\ &= \boldsymbol{G}\Delta\boldsymbol{q}\end{aligned} \tag{20}$$

where $\boldsymbol{G}$ is the error matrix, and $\boldsymbol{I}_{3\times3}$ is the unit matrix of 3×3.

3.3. Error model based on the double standard balls with distance constraint

The difference between the actual and predicted distances between two points in the working area of an industrial robot is known as the distance error [23]. Considering that the base coordinate system $\{O\}$ serves as the destination point's reference coordinate system. The schematic of error modeling based on the distance constraint is illustrated in Fig. 4, where A_i and B_i denote the spherical center points of the two standard balls described under the base coordinate system, and the distances L_t and L_r represent the theoretical and measured actual spherical center distances of the two standard balls, respectively.

$$L_t = \overline{p_{Ai}p_{Bi}} = \sqrt{(x_{Ai}-x_{Bi})^2 + (y_{Ai}-y_{Bi})^2 + (z_{Ai}-z_{Bi})^2} \tag{21}$$

Under the influence of errors $\Delta\boldsymbol{p}_{Ai}$ and $\Delta\boldsymbol{p}_{Bi}$, the theoretical spherical center points p_{Ai} and p_{Bi} result in the values of the actual measured points p'_{Ai} and p'_{Bi} are given in Eq. (22).

$$\begin{cases}\Delta\boldsymbol{p}_{Ai} = p'_{Ai} - p_{Ai} \\ \Delta\boldsymbol{p}_{Bi} = p'_{Bi} - p_{Bi}\end{cases} \tag{22}$$

The distance error $\Delta L_{i(AB)}$ between two measurement points (p_{Ai},p_{Bi}) is expressed in Eq. (23).

$$\overrightarrow{\Delta L_{i(AB)}} = \overline{p_{Ai}p_{Bi}} - \overline{p'_{Ai}p'_{Bi}} = \overrightarrow{L_t} - \overrightarrow{L_r} \tag{23}$$

$$\overrightarrow{L_t} = \overrightarrow{L_r} + \overrightarrow{\Delta L_{i(AB)}} = \overrightarrow{p_{Ai}} - \overrightarrow{p_{Bi}} \tag{24}$$

The expression of $\Delta L_{i(AB)}$ can be obtained from the relationship between error vectors [26].

$$\Delta L_{i(AB)} = \frac{[p_{Ai}-p_{Bi}]}{L_r}(\Delta\boldsymbol{p}_{Bi} - \Delta\boldsymbol{p}_{Ai}) \tag{25}$$

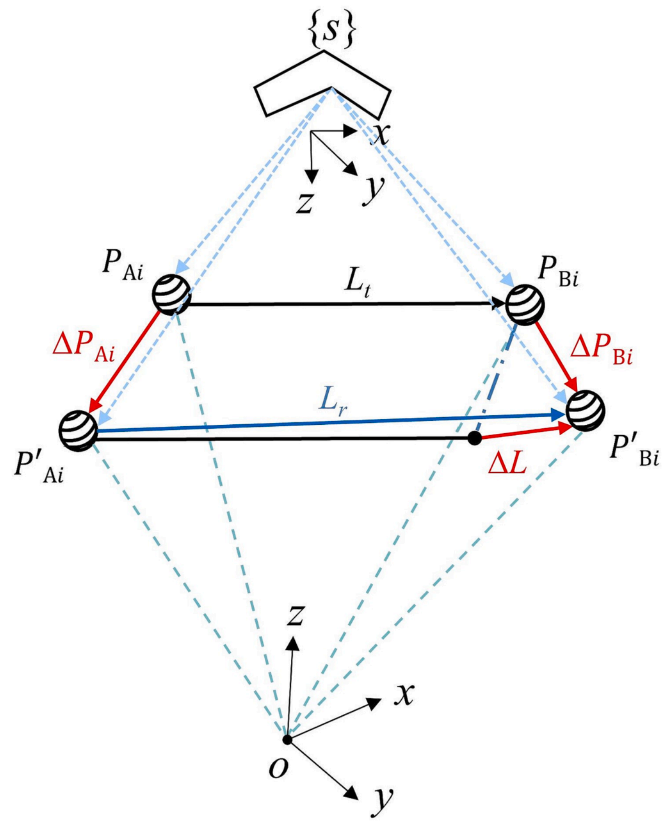

Fig. 4. Definition of robot distance error.

The relationship between distance error and measurement point error is obtained by substituting Eq. (20) into Eq. (25).

$$\begin{aligned}\Delta L_{i(AB)} &= \frac{[p_{Ai}-p_{Bi}]}{L_r}(\boldsymbol{G}_{Bi} - \boldsymbol{G}_{Ai})\Delta\boldsymbol{q} \\ &= \boldsymbol{H}_i\Delta\boldsymbol{q}\end{aligned} \tag{26}$$

Measurements on multiple sets of standard balls were performed to obtain error equations that can be used for parameter identification, as shown in Eq. (27).

$$\begin{bmatrix}\boldsymbol{H}_{B1} - \boldsymbol{H}_{A1} \\ \vdots \\ \boldsymbol{H}_{Bn} - \boldsymbol{H}_{An}\end{bmatrix}\Delta\boldsymbol{q} = \begin{bmatrix}\Delta L_{1(AB)} \\ \vdots \\ \Delta L_{n(AB)}\end{bmatrix} \tag{27}$$

3.4. Error parameter identification

The LM algorithm combines the benefits of the gradient descent method and the Gauss-Newton algorithm [24]. Based on the distance error model Eq. (27), the least-squares (LS) objective function $F(q)$ is built, and the optimal values of the hand-eye and kinematic parameter deviations are found using the LM algorithm.

$$F(\boldsymbol{q}) = \min\left(\frac{1}{2}f(\boldsymbol{q})^T f(\boldsymbol{q})\right) \tag{28}$$

where $\boldsymbol{q}$ is a vector containing the robot's hand-eye error and kinematic error parameters.

$$f(\boldsymbol{q}) = \begin{bmatrix}\Delta L_{1(AB)} & \Delta L_{2(AB)} & \cdots & \Delta L_{n(AB)}\end{bmatrix}^T \tag{29}$$

The above equation is expanded by Taylor series at $\boldsymbol{q}_i$, and the iterative Jacobian matrix $\boldsymbol{J}$ is constructed.

$$\boldsymbol{J}(\boldsymbol{q}_n) = \frac{\partial f(\boldsymbol{q}_n)}{\partial \boldsymbol{q}_i} \tag{30}$$

The recurrence formula of the LM algorithm is:

$$\Delta \boldsymbol{q}_n = -\left\{\left[\boldsymbol{J}(\boldsymbol{q}_n)^T\right]\boldsymbol{J}(\boldsymbol{q}_n) + \lambda \boldsymbol{I}\right\}^{-1}\boldsymbol{J}(\boldsymbol{q}_n)^T f(\boldsymbol{q}_n) \tag{31}$$

where $\Delta \boldsymbol{q}_n$ is the corrected value of the parameter error vector at the n_{th} iteration, $\boldsymbol{I}$ is the unit matrix, and λ is the damping factor.

After each iteration, the error parameter values are updated, and the equation $f(\boldsymbol{q})$ and Jacobian matrix $\boldsymbol{J}$ are calculated. When the stop condition is met, or the set number of iterations is completed, the cycle ends.

$$\boldsymbol{q}_{n+1} = \boldsymbol{q}_n + \Delta \boldsymbol{q}_n \tag{32}$$

If the error model's Jacobian matrix is linearly coupled, certain values may become unidentifiable, resulting in poor recognition accuracy. To avoid parameter redundancy, the robot error model should be made to retain only the necessary error parameters to ensure accurate calibration of the KUKA KR210 robot error. The redundant parameters corresponding to the elements that are zero on the diagonal of the matrix $\boldsymbol{R}$ are removed from the Jacobian matrix using the QR decomposition method. Through the above processing, the redundant parameters corresponding to the robot simultaneous calibration error model are a_1, a_2, α_1, α_3, d_2, z, and the 24 identifiable error parameters are obtained: a_3, a_4, a_5, a_6, α_2, α_4, α_5, α_6, d_1, d_3, d_4, d_5, d_6, θ_1, θ_2, θ_3, θ_4, θ_5, θ_6, x, y, A, B, C.

4. Error calibration and compensation experiment

4.1. Simultaneous calibration experiment

The major components of the KUKA robot hand-eye error and kinematic error simultaneous calibration platform are a six-degree-of-freedom KUKA KR210 industrial robot, an industrial control computer (IPC610I7-3770), a sensor bracket, a line-structured light sensor (Keyence LJ-G200), a fixture table, and two standard balls, as shown in Fig. 5.

In this section, simultaneous calibration experiments for hand-eye error and kinematic error were performed, and the accuracy of the calibration was verified. First, the hand-eye relationship matrix's initial values are calculated using the hand-eye initial value calibration scheme described in Section 3.1. Then, the simultaneous calibration experiment is carried out using the simultaneous calibration and identification approach, as presented in Sections 3.3 and 3.4. The experimental process is shown in Fig. 6, and the specific experimental steps are as follows.

Step 1: Fix a single standard ball at any position on the fixture table. Firstly, the robot is controlled to move in a way that only translates but does not rotate. The standard ball is measured, and its center coordinates are fitted to solve the rotation matrix $\boldsymbol{R}$. Then the robot moves to measure the standard ball in any attitude to solve the position vector $\boldsymbol{t}$.

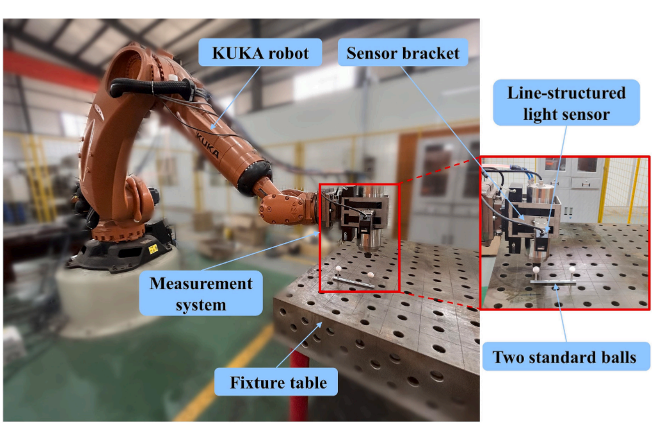

Fig. 5. Robot error calibration experimental platform.

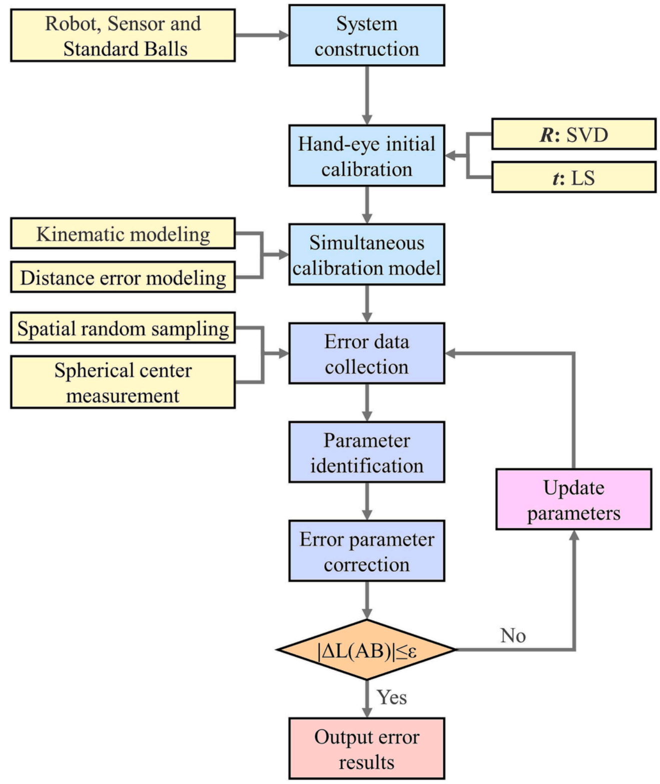

Fig. 6. Simultaneous calibration process of hand-eye and kinematic error.

Step 2: The midpoint of the line connecting the centers of the two standard balls is marked as M, and the actual distance L_r between them is measured using a coordinate measuring machine. In the robot operation space, a series of midpoints M_i are chosen randomly, as shown in Fig. 7.

Step 3: The robot is controlled to walk through each target point in turn to obtain multiple sets of joint angles. The robot stops for 30 seconds at each standard ball so that the line-structured light sensor can measure the coordinates of the current standard ball's center.

Step 4: The two standard balls' actual distance and distance error are calculated according to Eq. (26). The joint angle values are obtained by the robot control system, and the ball center coordinate values are

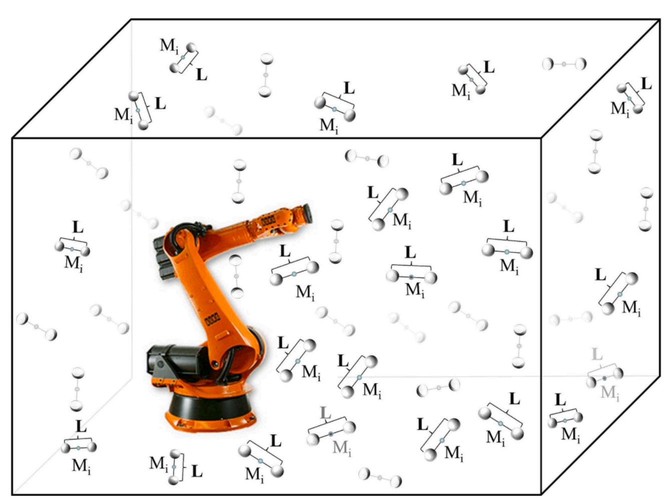

Fig. 7. Sampling point distribution map based on distance constraint.

obtained by the vision sensor. The resulting data are entered into Eq. (27), where the LM algorithm is used to identify the error values of each parameter.

Step 5: The robot's kinematic equations are corrected using the hand-eye and kinematic errors obtained in step 4, and the compensated angle values for each joint are calculated using inverse kinematics.

Step 6: The robot is controlled to reach the compensated target point to measure the current ball center coordinate value, and the distance error is calculated.

Step 7: Compare and analyze the distance error values before and after compensation.

The initial value of the hand-eye relationship obtained from step 1 is:

$$ {}^{m}_{s}\boldsymbol{R} = \begin{bmatrix} 0.2934 & -0.4449 & -0.8461 & 269.26 \\ 0.1271 & -0.8591 & 0.4958 & -97.05 \\ -0.9475 & -0.2530 & -0.1955 & 217.96 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

Tables 2 and 3 show the error results obtained by LS algorithm identification. The error parameter values obtained by the LS method are used as the initial values of the LM identification algorithm and then calibrated again. The results are shown in Tables 4 and 5. The calibration results can not be directly verified due to a lack of highly accurate measurement equipment. This research uses the distance error after error compensation as the evaluation index to validate the calibration accuracy.

4.2. Error compensation experiment

Error compensation refers to correcting the nominal kinematic parameters with the identified parameter errors and then modifying the kinematic parameters in the robot controller. The D-H parameters of the industrial robot control system cannot be altered directly due to the limitation of the system permission, so the positional errors in the cartesian space of the robot can only be compensated offline. The compensated joint angle values are obtained by substituting the corrected error parameter values into the robot's inverse kinematic equation [25–26], which are then fed into the control system to move the robot to the compensated position. Use the compensated angle value as the robot's operating command for accuracy verification experiments.

The principle of the direct compensation method is shown in Fig. 8. When the robot is compensated with Δx, due to the deviation of the theoretical kinematic parameters, there is still a deviation $\Delta x'$ in the joint variables. To improve the accuracy of error compensation, the iterative error compensation method is used in this paper. The remaining error Δx_i from the previous round is corrected using the iterative compensation method [27], which repeats the simultaneous calibration method in Section 4.1. The iteration stops when the error margin Δx_n meets the actual accuracy requirements ($|\Delta L_{(AB)}|<2.5$ mm). The principle of iterative error compensation is shown in Fig. 9. In this research, direct and indirect compensation experiments were carried out separately to compare the compensation effects.

The compensation of kinematic errors can only be achieved by compensating the joint angles, so it is necessary to calculate its compensation angle. Firstly, according to the corrected values of the robot hand-eye parameters and kinematic parameters, a new relationship matrix can be obtained, as shown in Eq. (33).

$$ {}^{B}_{s}\boldsymbol{T}^{new}_{(\theta)} = {}^{B}_{m}\boldsymbol{T}^{new}_{(\theta)}\,{}^{m}_{s}\boldsymbol{T}^{new} \tag{33} $$

where ${}^{B}_{s}\boldsymbol{T}^{new}_{(\theta)}$ is the transformation matrix from the modified visual coordinate system $\{s\}$ to the base coordinate system $\{B\}$, ${}^{B}_{m}\boldsymbol{T}^{new}_{(\theta)}$ is the transformation matrix from the modified robot end coordinate system $\{m\}$ to the base coordinate system $\{B\}$, and ${}^{m}_{s}\boldsymbol{T}^{new}$ is the modified hand-eye relationship matrix.

Assume that the amount of error in the target position P is $\boldsymbol{dp}$. According to Eq. (33), the compensated position ${}^{s}\boldsymbol{p}^{new}$ can be obtained by the following equation.

$$ {}^{s}\boldsymbol{p}^{new} = {}^{s}\boldsymbol{p}^{old} - \boldsymbol{dp} \tag{34} $$

where ${}^{s}\boldsymbol{p}^{old}$ is the position coordinate before compensation.

After obtaining the actual position coordinates, the joint angle value $\{\theta_1, \ldots, \theta_6\}$ is solved by inverse kinematics.

There is a small deviation from $\boldsymbol{dp}$ in the solution of Eq. (34), and the remaining small deviation needs to be calculated by Newton-Raphson

Table 2
D-H error by least squares identification.

Joint(i)	$a_{(i-1)}$/mm	$\alpha_{(i-1)}$/°	d_i/mm	θ_i/°
1	–	–	−0.579	−0.343
2	–	−0.406	–	−0.352
3	0.262	–	−0.103	0.295
4	0.403	0.176	0.547	−0.206
5	0.214	0.497	−0.098	−0.181
6	−0.196	0.101	−0.201	0.414

Table 3
Hand-eye error by least squares identification.

x/mm	y/mm	z/mm	A/rad	B/rad	C/rad
1.68	−0.635	–	0.00879	0.01204	0.00386

Table 4
D-H error by LM identification.

Joint(i)	$a_{(i-1)}$/mm	$\alpha_{(i-1)}$/°	d_i/mm	θ_i/°
1	–	–	−0.614	−0.741
2	–	−0.392	–	−0.411
3	0.362	–	−0.096	0.345
4	0.841	0.125	0.841	−0.098
5	0.161	0.628	0.164	−0.181
6	−0.271	0.051	−0.391	0.743

Table 5
Hand-eye error by LM identification.

x/mm	y/mm	z/mm	A/rad	B/rad	C/rad
1.35	−0.835	–	0.01663	0.01081	0.00431

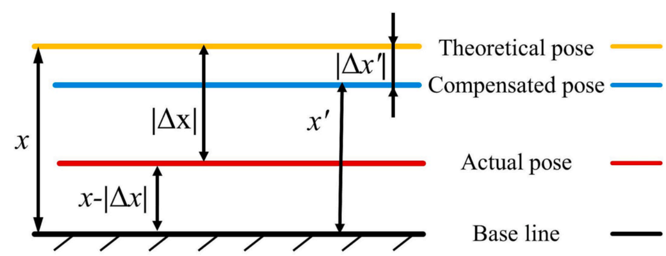

Fig. 8. The principle of direct error compensation.

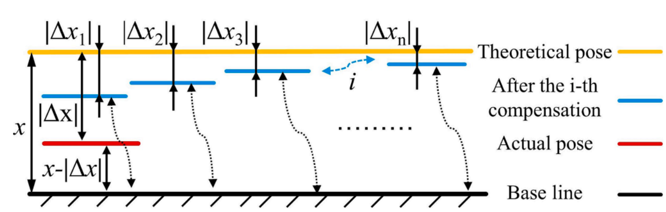

Fig. 9. The principle of iterative error compensation.

iteration.

$$ {}^{s}\boldsymbol{p}^{new} - {}^{s}\boldsymbol{p}^{old} = \frac{\partial\, {}^{s}\boldsymbol{p}^{old}}{\partial\theta_1}\theta_1^{'}\cdots\frac{\partial\, {}^{s}\boldsymbol{p}^{old}}{\partial\theta_n}\theta_n^{'} \tag{35} $$

The above partial differential equation is solved to obtain the value of $\{\theta_1^{'}, \ldots, \theta_6^{'}\}$, which is added to the angle value $\{\theta_1, \ldots, \theta_6\}$ to obtain the compensated robot joint angle.

4.3. Experimental results

An experimental study was carried out using the constructed experimental platform to verify the accuracy of the error parameter calibration. The error parameter identification algorithm and the error compensation algorithm were assessed by contrasting the distance error values before and after compensation.

In the robot's workspace, 60 sets of standard ball positions were randomly selected, and the center distance of the standard balls was measured using a line-structured light sensor, and the absolute values of the distance errors before and after calibration and compensation by different algorithms were counted. Robustness to uncertain parameters [28] and environmental variations [29] is one of the most important metrics for evaluating the performance of a perceptual measurement method. The line-structured light sensor used is experimentally verified to adapt to disturbing situations such as illumination changes and target shape changes, which can ensure the accuracy and reliability of the measurement results. Using the direct compensation approach, the identification accuracy of the LS and LM algorithms was evaluated, and the results are displayed in Fig. 10a. Further, the iterative compensation method proposed in 4.2 was verified based on the identification results of the LM algorithm. The results after iterative error compensation are shown in Fig. 10b.

Through Fig. 10, it is obvious that the positioning accuracy of the robot is significantly improved after the calibration using the simultaneous calibration method proposed in this paper. In order to further evaluate the experimental effect, the maximum error, average error, and standard deviation were analyzed according to statistical methods, and the results are shown in Table 6. The initial maximum distance error of the robot was 6.246 mm, and the average distance error was 3.967 mm. After calibration with LS and LM algorithms, the maximum distance error has been reduced to 3.854 mm and 3.142 mm, the average distance error has reduced to about 1.788 mm and 1.407 mm, and the positioning accuracy has been improved by 54.9% and 64.5%. The average distance error of the robot identified by the LM algorithm and compensated by iteration has been reduced to 1.001 mm, and the positioning accuracy has been improved by 74.8%. The standard deviation of the statistics in Table 6 can reflect the fluctuation range of the distance error value in Fig. 10. It can be seen that the error fluctuation of the LM algorithm combined with the iterative compensation method is the smallest, and the standard deviation is 0.396 mm. After using the synchronous calibration method proposed above, the positioning accuracy of the robot is greatly improved. In contrast, the LM algorithm combined with iterative compensation resulted in the greatest improvement in positioning accuracy and the smallest range of distance error fluctuations after robot calibration. Therefore, the identification results of the LM algorithm were used to iteratively compensate the robot positioning error and to guide the subsequent weld seams grinding of structural parts.

Table 6
Statistical analysis of distance error.

Distance error	Maximum error /(mm)	Average error /(mm)	Standard deviation /(mm)	Average error increased accuracy /(%)
No compensation	6.246	3.967	1.212	–
LS + direct compensation	3.854	1.788	0.848	54.9
LM + direct compensation	3.142	1.407	0.623	64.5
LM + iterative compensation	2.439	1.001	0.396	74.8

5. Conclusion

Error calibration is an effective means to improve robot positioning accuracy. Aiming at the problem of error accumulation caused by the traditional hand-eye error calibration process without considering the kinematic error, this paper proposes an error calibration method based on a line-structured light sensor. The method is able to synchronously calibrate the robot's hand-eye error and kinematic error, aiming at eliminating the error accumulation caused by the step-by-step calibration of hand-eye error and kinematic error and guiding the subsequent weld seams grinding to improve the robot's grinding accuracy. The main contents and conclusions are as follows.

1) Based on the principle of differential kinematics, the simultaneous calibration error model was derived, and the robot position error was converted into distance error for the solution. The LS and LM algorithms were used to identify error parameters, and the error was compensated by the direct compensation method and iterative compensation method.

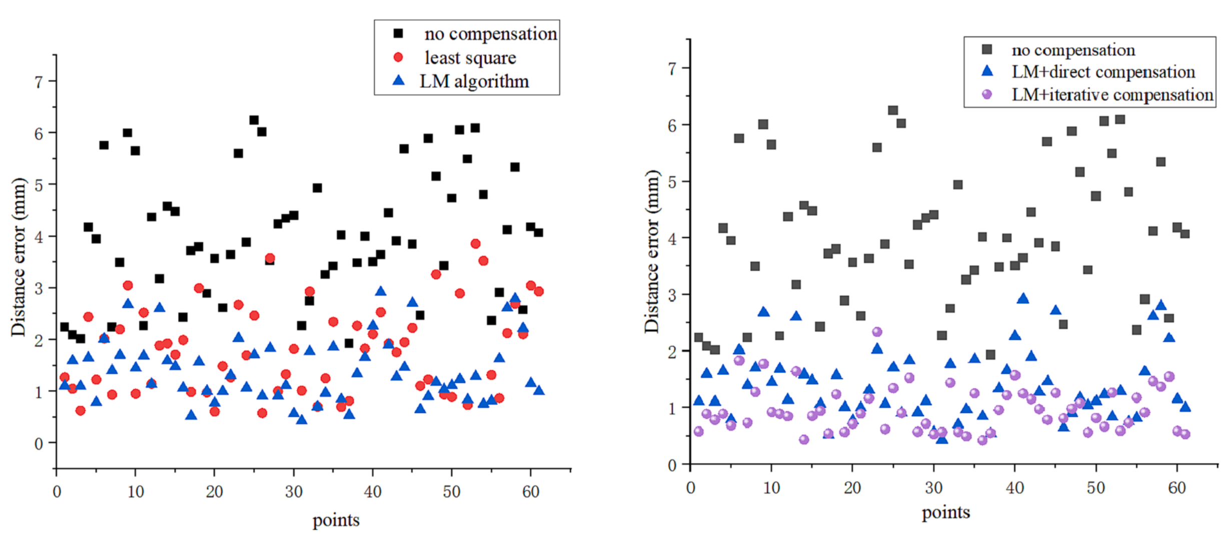

Fig. 10. Distance error comparison before and after calibration.

2) A synchronous calibration experimental platform based on a line-structured light sensor was built, and the calibration accuracy was experimentally verified by measuring the center distance of two standard balls with known distances. Through the statistics, analysis, and comparison of the distance error data before and after calibration, the LM algorithm for identification and the iterative approach for compensating can better increase the positioning accuracy of the robot. The experimental results show that the proposed method can effectively calibrate the robot's hand-eye and kinematic parameters, and the average distance error has been reduced by 74.8%.

The present work only calibrated the kinematic error of the robot without considering the non-kinematic error. When the robot is used for high-precision manufacturing, the non-kinematic errors caused by the stiffness of the connecting rod or the flexibility of the joint should be further considered. In the next stage, the flexibility error caused by the load will be compensated and integrated into the robot grinding and polishing system.

CRediT authorship contribution statement

Dahu Cao: Writing – original draft, Conceptualization, Methodology, Formal analysis. **Wei Liu:** Conceptualization, Methodology, Investigation, Resources, Writing – review & editing. **Shun Liu:** Conceptualization, Methodology, Data curation, Writing – review & editing. **Jia Chen:** Investigation, Validation. **Wang Liu:** Investigation, Validation. **Jimin Ge:** Investigation, Validation. **Zhaohui Deng:** Supervision, Resources.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] M.R. Pedersen, L. Nalpantidis, R.S. Andersen, C. Schou, S. Bøgh, V. Krüger, O. Madsen, Robot skills for manufacturing: From concept to industrial deployment, Rob. Comput. Integr. Manuf. 37 (2016) 282–291, https://doi.org/10.1016/j.rcim.2015.04.002.
[2] Y.K. Liu, Y.M. Zhang, Fusing machine algorithm with welder intelligence for adaptive welding robots, J. Manuf. Process. 27 (2017) 18–25, https://doi.org/10.1016/j.jmapro.2017.03.015.
[3] I. Iglesias, M.A. Sebastián, J.E. Ares, Overview of the state of robotic machining: current situation and future potential, Procedia. Eng. 132 (2015) 911–917, https://doi.org/10.1016/j.proeng.2015.12.577.
[4] G. Michalos, N. Kousi, P. Karagiannis, C. Gkournelos, K. Dimoulas, S. Koukas, K. Mparis, A. Papavasileiou, S. Makris, Seamless human robot collaborative assembly-An automotive case study, Mechatronics 55 (2018) 194–211, https://doi.org/10.1016/j.mechatronics.2018.08.006.
[5] Y.B. Song, W.J. Tian, Y.L. Tian, X.P. Liu, A task-oriented calibration method for a 5-DOF hybrid machining robot based on the equivalent system, Measurement 216 (2023), 112909, https://doi.org/10.1016/j.measurement.2023.112909.
[6] J.M. Ge, Z.H. Deng, Z.Y. Li, T. Liu, R.J. Zhuo, X. Chen, Adaptive parameter optimization approach for robotic grinding of weld seam based on laser vision sensor, Rob. Comput. Integr. Manuf. 82 (2023), https://doi.org/10.1016/j.rcim.2023.102540.
[7] Z.R. Zhu, X.W. Tang, C. Chen, F.Y. Peng, R. Yan, L. Zhou, Z.P. Li, J.W. Wu, High precision and efficiency robotic milling of complex parts: Challenges, approaches and trends, Chin. J. Aeronaut. 35 (2022) 22–46, https://doi.org/10.1016/j.cja.2020.12.030.
[8] Y. Wu, A. Klimchik, S. Caro, B. Furet, A. Pashkevich, Geometric calibration of industrial robots using enhanced partial pose measurements and design of experiments, Rob. Comput. Integr. Manuf. 35 (2015) 151–168, https://doi.org/10.1016/j.rcim.2015.03.007.
[9] Y.X. Guo, Z.X. Jiang, B. Song, X.Q. Tang, H.S. Min, C.L. Fu, A distance calibration method for kinematic parameters of serial industrial robots considering the accuracy of relative position, Measurement 204 (2022), 111842, https://doi.org/10.1016/j.measurement.2022.111842.
[10] A. Nubiola, I.A. Bonev, Absolute robot calibration with a single telescoping ballbar, Precis. Eng. 38 (2014) 472–480, https://doi.org/10.1016/j.precisioneng.2014.01.001.
[11] X.Y. Chen, Q.J. Zhang, Y.L. Sun, Non-kinematic calibration of industrial robots using a rigid-flexible coupling error model and a full pose measurement method, Rob. Comput. Integr. Manuf. 57 (2019) 46–58, https://doi.org/10.1016/j.rcim.2018.07.002.
[12] J. Santolaria, M. Ginés, Uncertainty estimation in robot kinematic calibration, Rob. Comput. Integr. Manuf. 29 (2) (2013) 370–384, https://doi.org/10.1016/j.rcim.2012.09.007.
[13] C. Yang, W.Z. Lu, Y.Q. Xia, Positioning accuracy analysis of industrial robots based on non-probabilistic time-dependent reliability, Ieee T Reliab. (2023) 1–14, https://doi.org/10.1109/TR.2023.3292089.
[14] A. Nubiola, I.A. Bonev, Absolute calibration of an ABB IRB 1600 robot using a laser tracker, Rob. Comput. Integr. Manuf. 29 (2013) 236–245, https://doi.org/10.1016/j.rcim.2012.06.004.
[15] Y.P. Tang, W.X. Luo, Q. Wang, J.Y. Li, L. Cheng, J.X. Li, Y.L. Ke, A novel hand-eye semi-automatic calibration process for laser profilometers using machine learning, Measurement 216 (2023), 112936, https://doi.org/10.1016/j.measurement.2023.112936.
[16] S.P. Zhang, Y.S. Ding, K.R. Hao, D. Zhang, An efficient two-step solution for vision-based pose determination of a parallel manipulator, Rob. Comput. Integr. Manuf. 28 (2012) 182–189, https://doi.org/10.1016/j.rcim.2011.08.005.
[17] Y.L. Kuo, B.H. Liu, C.Y. Wu, Pose determination of a robot manipulator based on monocular vision, IEEE Access 4 (2016) 8454–8464, https://doi.org/10.1109/access.2016.2633378.
[18] Y.C. Shiu, S. Ahmad, Calibration of wrist-mounted robotic sensors by solving homogeneous transform equations of the form AX=Xb, Ieee. T. Autom. Sci. Eng. 5 (1) (1989) 16–29, https://doi.org/10.1109/70.88014.
[19] I. Enebuse, M. Foo, B.S.K.K. Ibrahim, H. Ahmed, F. Supmak, O.S. Eyobu, A comparative review of hand-eye calibration techniques for vision guided robots, IEEE Access 9 (2021) 113143–113155, https://doi.org/10.1109/access.2021.3104514.
[20] D. Condurache, A. Burlacu, Orthogonal dual tensor method for solving the AX=XB sensor calibration problem, Mech. Mach. Theory. 104 (2016) 382–404, https://doi.org/10.1016/j.mechmachtheory.2016.06.002.
[21] J. Heller, M. Havlena, T. Pajdla, Globally optimal hand-eye calibration using branch-and-bound, Ieee. Trans. Pattern. Anal. Mach. Intell. 38 (2016) 1027–1033, https://doi.org/10.1109/TPAMI.2015.2469299.
[22] Z.Y. Liu, X. Liu, G.F. Duan, J.R. Tan, Precise hand-eye calibration method based on spatial distance and epipolar constraints, Rob. Auton. Syst. 145 (2021), https://doi.org/10.1016/j.robot.2021.103868.
[23] X.C. Zhou, Q.X. Zhang, S.X. Zheng, A new model with compensation algorithm for distance errors of robot mechanisms, Robot 13 (1) (1991) 44–49. https://doi.org/10.13973/j.cnki.robot.1991.01.008.
[24] Q.D. Zhu, X.R. Xie, C. Li, G.H. Xia, Q. Liu, Kinematic self-calibration method for dual-manipulators based on optical axis constraint, IEEE Access 7 (2019) 7768–7782, https://doi.org/10.1109/access.2018.2890123.
[25] A.A. Goldenberg, D.L. Lawrence, A generalized solution to the inverse kinematics of robot manipulators, J. Dyn. Sys., Meas., Control. 107 (1) (1985) 103–106, https://doi.org/10.1115/1.3140699.
[26] J. Li, H. Yu, N.Y. Shen, Z. Zhong, Y.H. Lu, J.C. Fan, A novel inverse kinematics method for 6-DOF robots with non-spherical wrist, Mech. Mach. Theory. 157 (2021), https://doi.org/10.1016/j.mechmachtheory.2020.104180.
[27] P. Dépincé, J.Y. Hascoët, Active integration of tool deflection effects in end milling, Part 2. Compensation of tool deflection, Int. J. Mach. Tool. Manu. 46 (2006) 945–956, https://doi.org/10.1016/j.ijmachtools.2005.08.014.
[28] C. Yang, W.Z. Lu, Y.Q. Xia, Reliability-constrained optimal attitude-vibration control for rigid-flexible coupling satellite using interval dimension-wise analysis, Reliab Eng Syst Safe. 237 (2023), 109382, https://doi.org/10.1016/j.ress.2023.109382.
[29] C. Yang, Y.Q. Xia, Interval uncertainty-oriented optimal control method for spacecraft attitude control, Ieee T Aero Elec Sys. (2023) 1–13, https://doi.org/10.1109/TAES.2023.3257777.